

Announcement

Homework #5 has been posted in Blackboard and course website

Due: 1:15pm, Thursday, April 9

Today's Agenda

- **Image Compression**

Fundamentals

Data represent information – different ways

Representations that contain irrelevant or repeated information → contain redundant data

Two representations of the same information: b and b' bits, then the relative data redundancy

$$R = 1 - \frac{1}{C}, \quad \text{where}$$

$$C = \frac{b}{b'} \text{ is called the compression ratio}$$

In digital image processing, b is the # bits for the 2D array representation and b' is the compressed representation

Three Types of Data Redundancies

Coding redundancy

- Code/code book is a system to represent information
- Code length is the number of symbols in each code word
- Do we really need 8 bits to represent a gray-level pixel?

Spatial and temporal redundancy

- Neighboring (spatially or temporally) pixels usually have similar intensities!
- Do we need to represent every pixel?

Irrelevant information

- Some image information can be ignored.

Coding Redundancy

Histogram

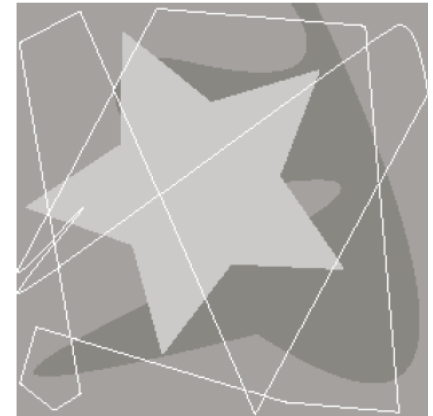
$$p_r(r_k) = \frac{n_k}{MN}, \quad k = 0, 1, 2, \dots, L-1$$

Average # bits required to represent each pixel

$$L_{\text{avg}} = \sum_{k=0}^{L-1} l(r_k) p_r(r_k)$$

Number of bits representing each intensity level

Total bits MNL_{avg}



r_k	$p_r(r_k)$	Code 1	$l_1(r_k)$	Code 2	$l_2(r_k)$
$r_{87} = 87$	0.25	01010111	8	01	2
$r_{128} = 128$	0.47	10000000	8	1	1
$r_{186} = 186$	0.25	11000100	8	000	3
$r_{255} = 255$	0.03	11111111	8	001	3
r_k for $k \neq 87, 128, 186, 255$	0	—	8	—	0

TABLE 8.1
Example of
variable-length
coding.

Fixed length

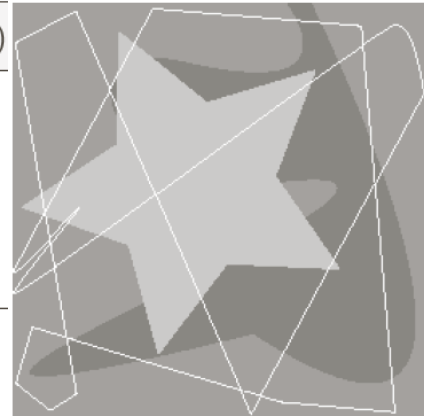
Variable length

$$L_{\text{avg}} = 2 * 0.25 + 0.47 * 1 + 0.25 * 3 + 0.03 * 3 = 1.8$$

Coding Redundancy

TABLE 8.1
Example of
variable-length
coding.

r_k	$p_r(r_k)$	Code 1	$l_1(r_k)$	Code 2	$l_2(r_k)$
$r_{87} = 87$	0.25	01010111	8	01	2
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r_k for $k \neq 87, 128, 186, 255$	0	—	8	—	0



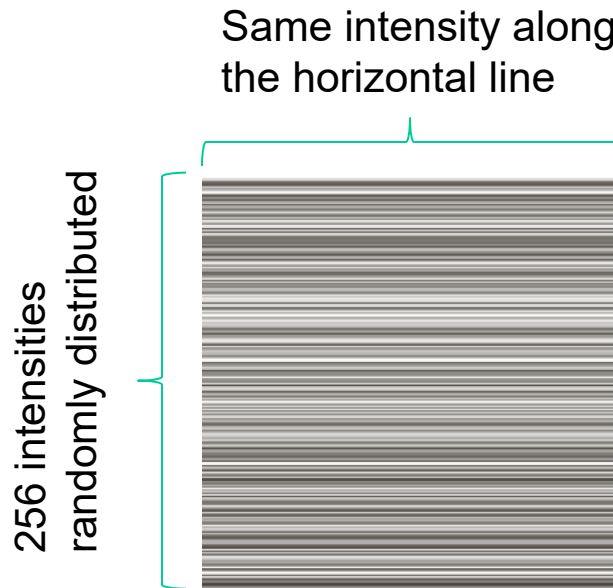
C and R with variable length coding?

$$C = \frac{8}{1.81} = 4.42$$

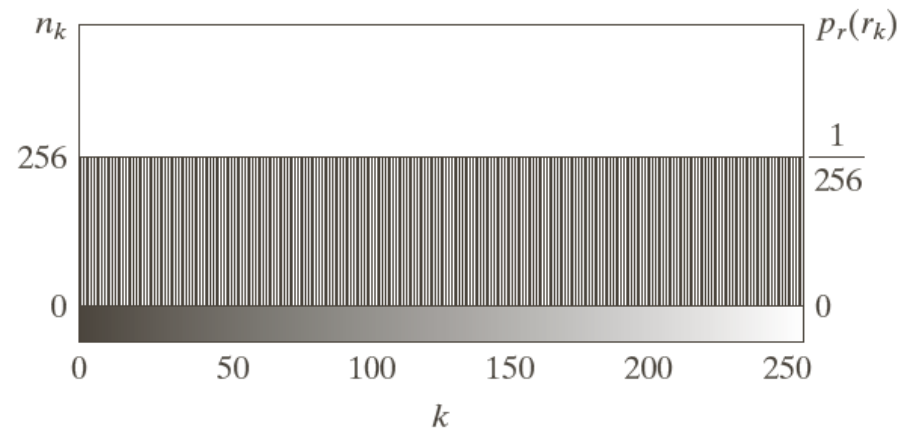
$$R = 1 - \frac{1}{C} = 0.77$$

Spatial and Temporal Redundancy

FIGURE 8.2 The intensity histogram of the image in Fig. 8.1(b).



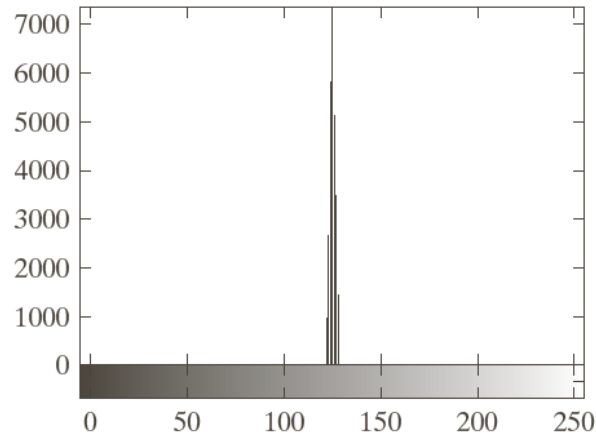
The histogram is uniform.



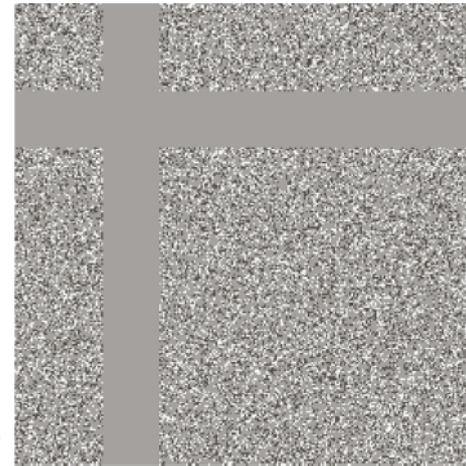
Compression by mapping:

- Run-length coding:
 - one word representing the intensity, and one word representing the length
 - C and R?
- Difference between two neighboring pixels

Irrelevant Information



Represented by a single byte



a b

FIGURE 8.3
(a) Histogram of the image in Fig. 8.1(c) and (b) a histogram equalized version of the image.

More details

**Quantization – loss of quantitative information:
irreversible operation**

Measuring Image Information

Minimum amount of data without losing information?

A random event E with probability $P(E)$ contains information

$$I = \log \frac{1}{P(E)} = -\log P(E)$$

Entropy (average information per image intensity)

$$H = -\sum_{k=0}^{L-1} p_r(r_k) \log_2 p_r(r_k)$$

Shannon's first theorem (noiseless coding theorem)

$$L_{\text{avg}} \geq H$$

and the low-bound H can be achieved by a coding method

Fidelity Criteria – Quantify the Loss

objective fidelity criteria

Root mean square error

$$e_{\text{rms}} = \left[\frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [\hat{f}(x, y) - f(x, y)]^2 \right]^{\frac{1}{2}}$$

Mean-square signal to noise ratio

$$SNR_{\text{ms}} = \frac{\sum_{x=0}^{M-1} \sum_{y=0}^{N-1} \hat{f}(x, y)^2}{\sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [\hat{f}(x, y) - f(x, y)]^2}$$

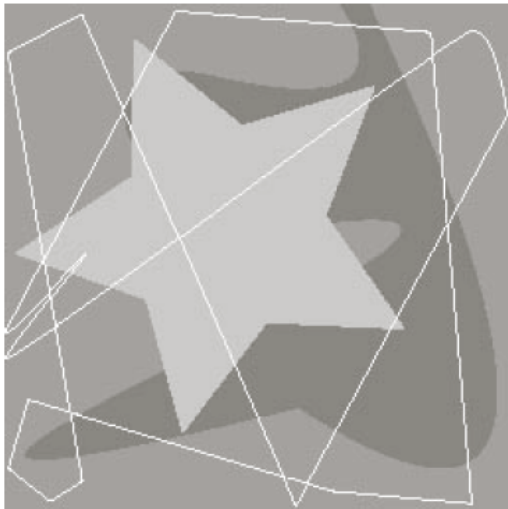
TABLE 8.2
 Rating scale of
 the Television
 Allocations Study
 Organization.
 (Freundtall and
 Behrend.)

Subjective Fidelity Criteria

Value	Rating	Description
1	Excellent	An image of extremely high quality, as good as you could desire.
2	Fine	An image of high quality, providing enjoyable viewing. Interference is not objectionable.
3	Passable	An image of acceptable quality. Interference is not objectionable.
4	Marginal	An image of poor quality; you wish you could improve it. Interference is somewhat objectionable.
5	Inferior	A very poor image, but you could watch it. Objectionable interference is definitely present.
6	Unusable	An image so bad that you could not watch it.

Inconsistency between Objective and Subjective Fidelity Criteria

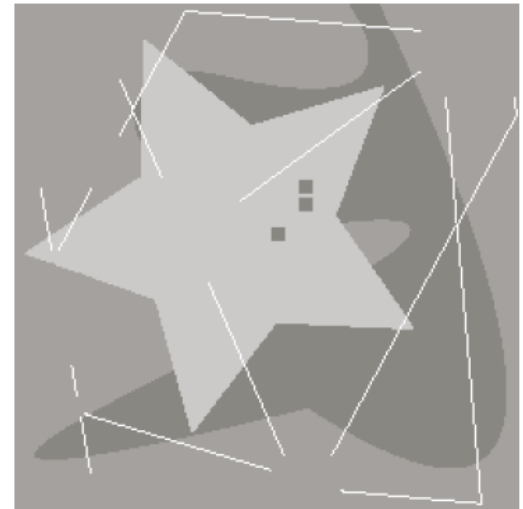
Objective: rms = 5.17



rms = 15.67



rms = 14.17



Subjective: Excellent

Passable

unusable

a b c

FIGURE 8.4 Three approximations of the image in Fig. 8.1(a).

Image-Compression Models

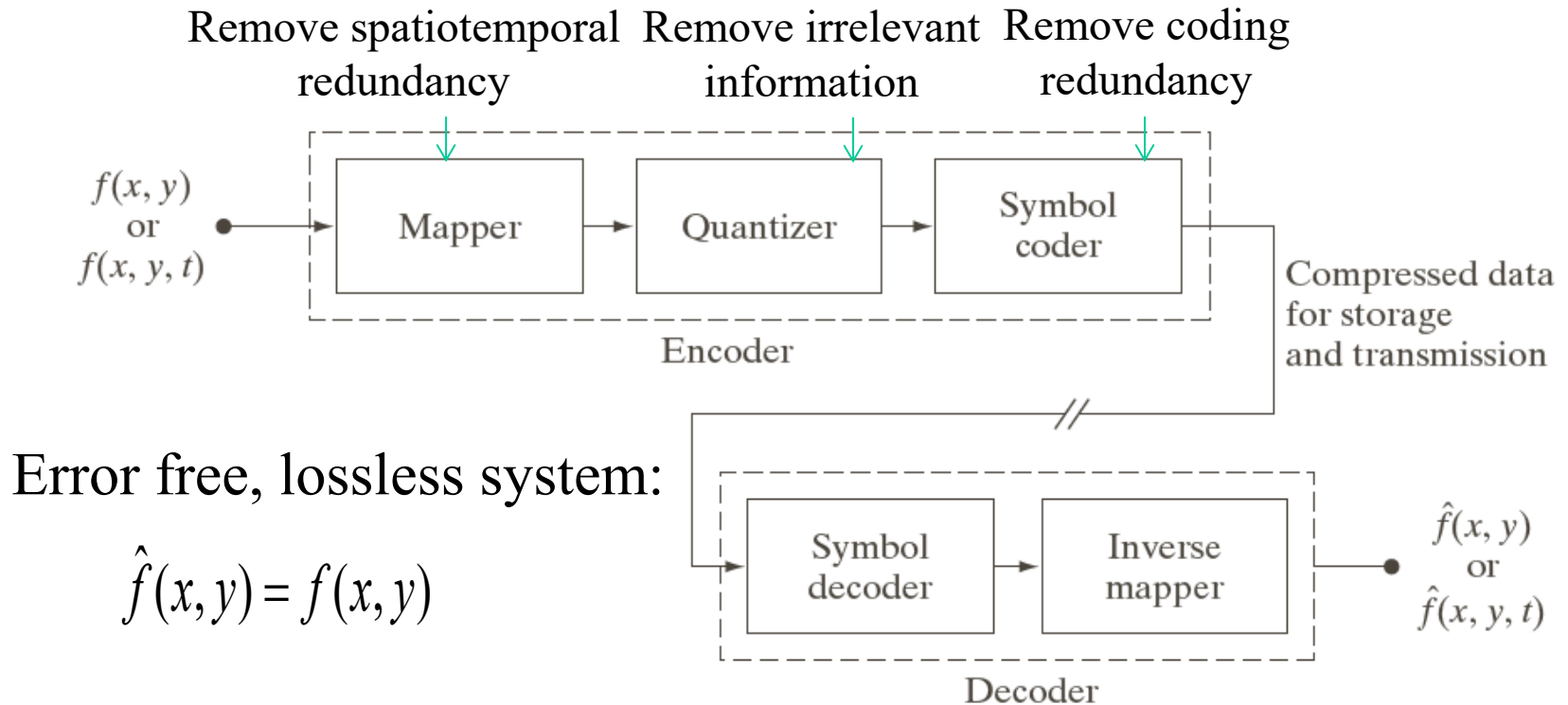


FIGURE 8.5
Functional block
diagram of a
general image
compression
system.

Image Formats, Containers and Compression Standards

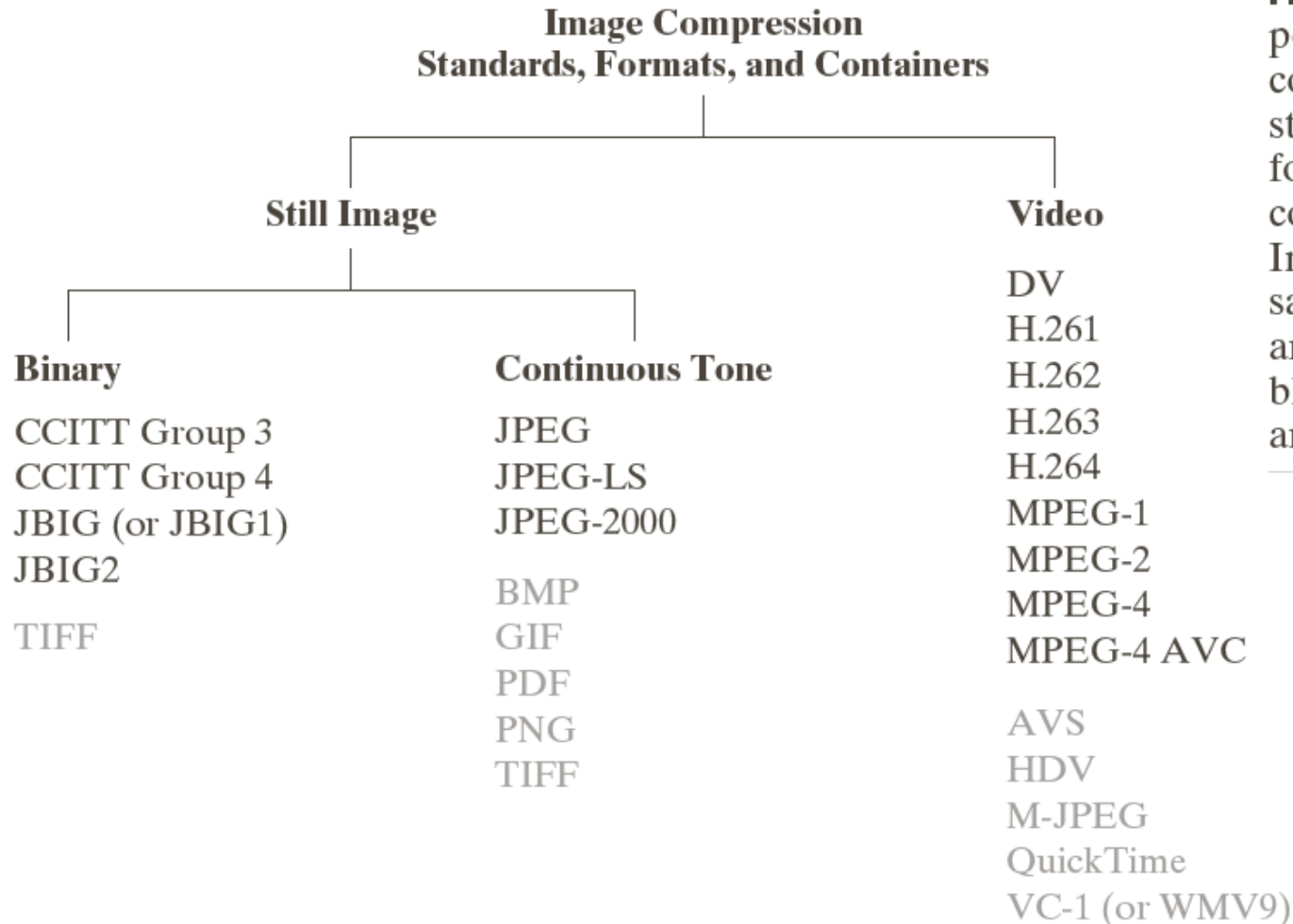


FIGURE 8.6 Some popular image compression standards, file formats, and containers. Internationally sanctioned entries are shown in black; all others are grayed.

Some Basic Compression Methods – Huffman Coding (Block Code)

- Remove coding redundancy
- Used widely in CCITT group3, JBIG2, JPEG, and MPEG
- Create the optimal code for a set of symbols

Original source		Source reduction			
Symbol	Probability	1	2	3	4
a_2	0.4	0.4	0.4	0.4	0.6
a_6	0.3	0.3	0.3	0.3	
a_1	0.1	0.1	0.2	0.3	0.4
a_4	0.1	0.1			
a_3	0.06	0.1	0.1	0.1	0.1
a_5	0.04				

FIGURE 8.7
Huffman source reductions.

Huffman Coding – Cont.

Original source			Source reduction			
Symbol	Probability	Code	1	2	3	4
a_2	0.4	1	0.4 1	0.4 1	0.4 1	0.6 0
a_6	0.3	00	0.3 00	0.3 00	0.3 00	0.4 1
a_1	0.1	011	0.1 011	0.2 010	0.3 01	
a_4	0.1	0100	0.1 0100	0.1 011		
a_3	0.06	01010	0.1 0101			
a_5	0.04	01011				

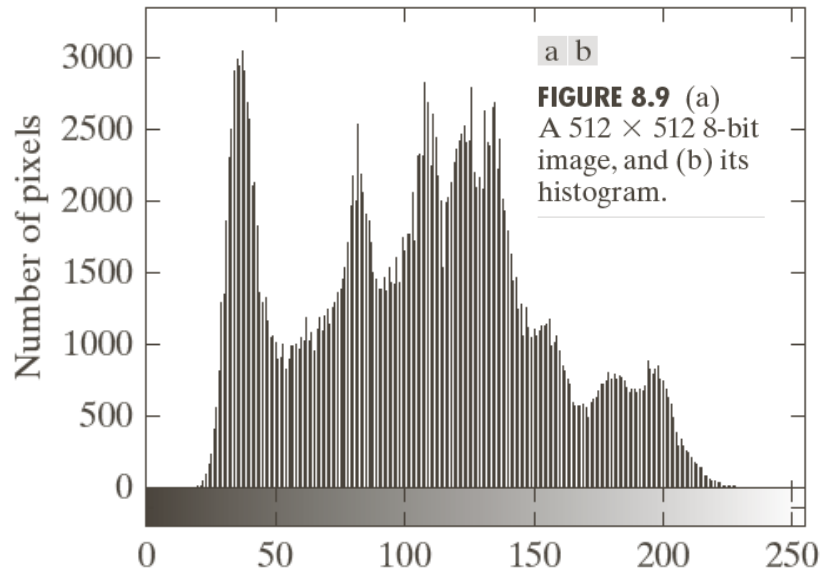
FIGURE 8.8
Huffman code
assignment
procedure.

$$L_{\text{avg}} = (0.4)(1) + (0.3)(2) + (0.1)(3) + (0.1)(4) + (0.06)(5) + (0.04)(5) = 2.2 \text{ bits/pixel}$$

$$H = 2.14 \text{ bits/pixel}$$

- **Block code:** each source symbol is represented by a fixed code symbol
- **Instantaneous:** lookup table
- **Uniquely decodable:** extract symbols in a left-to-right manner

Example



→ 7.428 bits/pixel

In practice, a pre-computed Huffman coding table is used (e.g., JPEG and MPEG)

Arithmetic Coding -- Nonblock Code

- Used in JBIG, JBIG2, JPEG2000, and MPEG4
- Non-block: the whole message is encoded into a single code word (real value in $[0, 1]$)

Source Symbol	Probability	Initial Subinterval
a_1	0.2	$[0.0, 0.2)$
a_2	0.2	$[0.2, 0.4)$
a_3	0.4	$[0.4, 0.8)$
a_4	0.2	$[0.8, 1.0)$

TABLE 8.6
Arithmetic coding
example.

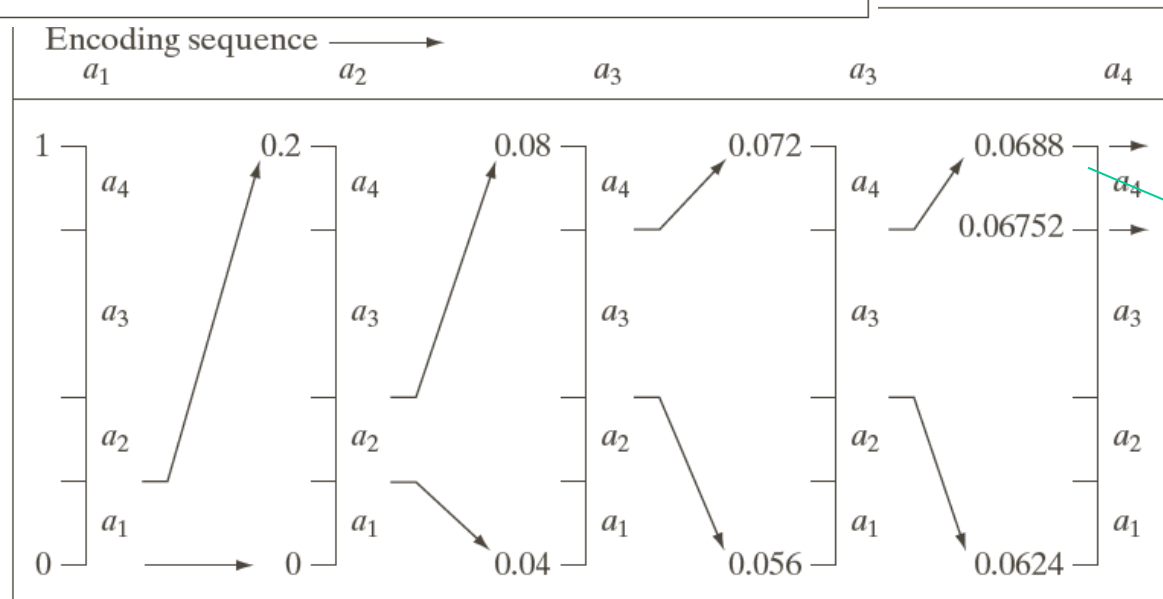


FIGURE 8.12
Arithmetic coding
procedure.

Any number in the
range can be used

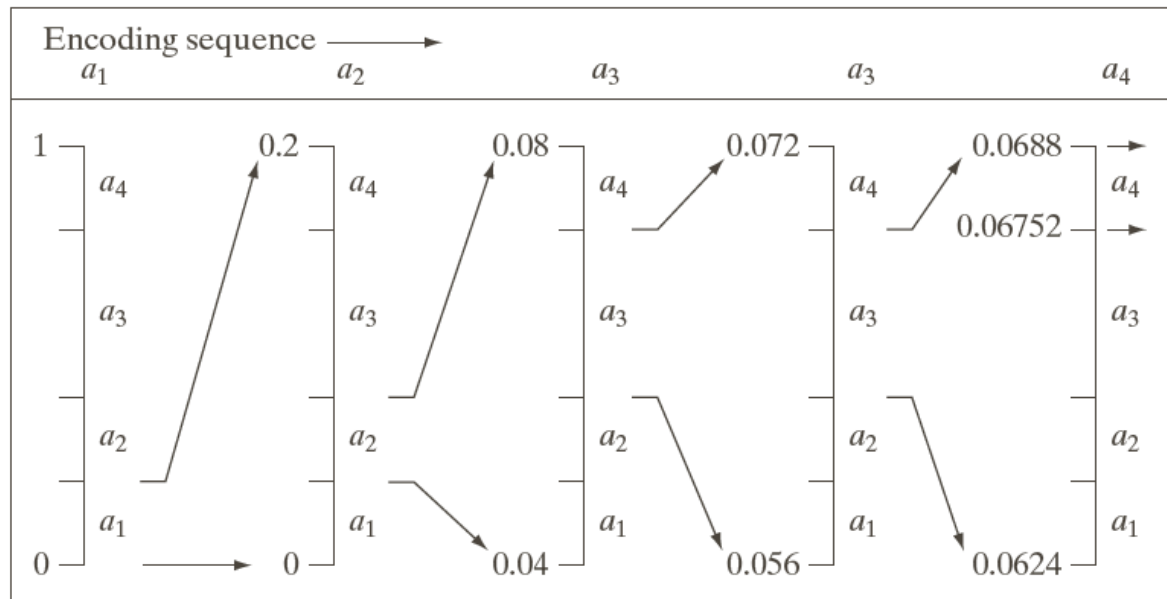
Arithmetic Coding

Decoding

- final value
- probabilities of the input symbols

Two decoding methods:

- Straightforward decoding



Arithmetic Coding

Decoding

- final value
- probabilities of the input symbols

Two decoding methods:

- Straightforward decoding
- An efficient method

Source Symbol	Probability	Initial Subinterval
a_1	0.2	[0.0, 0.2)
a_2	0.2	[0.2, 0.4)
a_3	0.4	[0.4, 0.8)
a_4	0.2	[0.8, 1.0)

Step0: $v_t = v_0$

Repeat:

step1: find symbol s_t satisfying $low(s_t) \leq v_t \leq up(s_t)$

step2:
$$v_{t+1} = \frac{v_t - low(s_t)}{p(s_t)}$$

Until: s_t is the end symbol

Arithmetic Coding

Require an end-of-message indicator

Potential issues:

- Decoding starts when all the message is received
- Sensitive to the noise during transmission
- Limited by the precision – solved by scaling

Run-Length Encoding (RLE)

- Eliminate spatial redundancy – groups of identical symbols
- Used in CCITT, JPEG, and fax machines

Encoded Mode: two bytes

of consecutive pixels, the color index

Absolute Mode: two bytes

0, conditional signals

Second Byte Value	Condition
0	End of line
1	End of image
2	Move to a new position
3–255	Specify pixels individually

TABLE 8.8

BMP absolute coding mode options. In this mode, the first byte of the BMP pair is 0.

Run-Length Encoding (RLE)



$C=0.98$



$C=1.35$



$C=128$

- Effective on binary images (black and white)
- Not effective on natural images - increase the file size if there are few runs of identical symbols
- Combine with other variable-length coding

Motion Compensation

Eliminate temporal redundancy

- Step1 : Capture differences between a reference frame and the target frame
- Step2 : Model the difference in terms of transformation function
- Step3 : Video/image sequences are synthesized by the reference frame according to the estimate transformation

Motion Compensation

Eliminate temporal redundancy

- Step1 : Capture differences between a reference frame and the target frame
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Global motion compensation

- Few parameters; no partition;
- Static objects; not applicable for moving objects

Local motion compensation

- Block motion compensation
- Overlapped block motion compensation

Reading Assignments

Other methods covered in Chapter 8.2